Dylan Atkins

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Research Portfolio: dylan-atkins-chemical-engineering.onrender.com

Education

Florida State University, Tallahassee, FL **Expected May 2029**
B.S. Chemical Engineering | GPA: 3.74/4.00

Palm Beach State College, Palm Beach Gardens/Lake Worth, FL **May 2026**
Associate of Arts

Technical Skills

Laboratory & Materials Science: Polymer synthesis (PAA–PVA networks), materials characterization, and mechanical testing; hydrogel fabrication and crosslinking; failure analysis; solution preparation; titration; pH measurement (LabQuest); experimental design and optimization; statistical analysis (t-tests).

Programming & Data: Microsoft Excel (data organization and analysis) and Python (scientific computing, data analysis, automation fundamentals); data visualization; experimental data analysis.

Engineering Design (CAD): Autodesk Inventor; Autodesk Revit; Onshape; Fusion 360; SolidWorks; engineering drawings and 3D modeling.

Biotechnology: DNA extraction; PCR; gel electrophoresis; DNA Subway & BLAST analysis; sterile technique; microbial culturing (Kirby–Bauer method).

Work & Research Experience

Shaw Industries – Plant 70, Bainbridge, GA **August 2026**
Chemistry Laboratory & Process Engineering Job Shadow
- Gained hands-on exposure to polymer crosslinking system optimization and quality control (QC) while shadowing a Chemist and Process Engineering team in synthesizing and characterizing composite materials.
- Observed process optimization and manufacturing operations that connect materials testing to production-scale reliability.

Independent Research Project **May 2025 – December 2025**
Optimization of Poly (acrylic acid) Polymer Networks for Extrusion and 3D Forming Applications
Supervisor(s): William Bartenslager (william.bartenslager@palmbeachschools.org)
- Developed low-temperature PAA–PVA hydrogel filaments as materials-engineered alternatives to petroleum-based 3D printing feedstocks.
- Conducted rigorous materials characterization, including mechanical testing, solvent stability analysis, and extrusion flow diagnostics, to evaluate material performance and failure behavior.
- Optimized crosslinking systems (borax, glycerol, CaCl2, NaPAA), achieving statistically validated improvements in tear force and extensibility (t-test validated).
- Designed PAA–PAH polyelectrolyte outer films to mitigate shear-thinning and enable continuous extrusion with partial self-healing.

Honors: First Place, School Science Fair (2025); Finalist, Palm Beach Regional Science Fair (2025)

Coastal Ocean Explorers: Shark Program **October 2025**
ANGARI Foundation & Florida International University
Supervisors: ANGARI Foundation; Florida International University
- Performed DNA extraction, PCR, and gel electrophoresis under sterile laboratory technique to identify shark prey species from field and market samples.
- Used DNA Subway to construct a phylogenetic tree and prepared results for conference presentation.

Independent Research Project **May 2024 – January 2025**
Effect of Aluminum Hydroxide on Lithium-Ion Separation
Supervisor(s): William Bartenslager
- **Investigated lithium separation** from lithium carbonate using aluminum hydroxide, examining how experimental conditions influenced separation outcomes.
- Conducted titrations, pH monitoring, mass analysis, and statistical comparisons to evaluate experimental results.
- Applied controlled experimental procedures and data analysis to investigate lithium recovery, building an interest in materials processing and energy-related technologies.

Honors: First Place, School Science Fair (2024); Third Place Chemistry & Finalist, Palm Beach Regional Science Fair (2024)

Independent Research Project **August 2023-December 2023**
Perfluorooctanoic Acid (PFOA) Removal from Water Using Dimethyl Sulfoxide and Granular Activated Carbon
Supervisor(s): William Bartenslager
- Designed a solvent-assisted adsorption approach for PFAS remediation using dimethyl sulfoxide and granular activated carbon.
- Validated adsorbent material performance through controlled GAC adsorption trials and mass comparison analysis.

Honors: First Place, School Science Fair (2023)

Awards & Honors

- QuestBridge National College Match Finalist (2025)
- Planting Science Star Project Awardee (2025)
- Palm Beach Regional Science Fair Finalist (2025)
- 1st Place, Palm Beach Central School Science Fair (2023–2025)
- 3rd Place, Chemistry, Palm Beach Regional Science Fair (2024)
- Citizen Science Participant (2025)
- Science Olympiad Participant (2025)
- Max Planck Brain Bee Participant (2025)